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# Intrusion-Resilient Intelligent Routing in VANETs Using Dynamic Optimization

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## Abstract

Efficient routing in Vehicular Ad Hoc Networks (VANETs) is crucial for ensuring reliable data transfer, extending network lifetime, and preserving energy. Existing methods often face challenges in energy efficiency and cyber security. This study aims to develop a novel Dynamic Waterwheel Plant Optimization (DWPO) strategy is integrated with blockchain to provide efficient path planning and enhanced cyber security in VANETs. The primary research question investigates whether DWPO outperform existing routing methods in terms of energy, latency, throughput, and security. The proposed DWPO approach

employs a fitness function that incorporates inter-cluster distance and other relevant network parameters to select optimal routing paths. Blockchain is integrated to secure routing information and reduce the risk of malicious attacks. The model is implemented and tested in MATLAB, and its performance is compared with existing approaches. Experimental results demonstrate that the DWPO model achieves a network lifetime of 93.6%, a packet delivery ratio of 90.2%, energy consumption of 18.2 J, end-to-end delay of 99.372 ms, throughput of 879.34 kbps, and energy efficiency of 83.72%. Compared to existing methods, DWPO reduces energy consumption, minimizes latency, increases data transmission capability, and maintains stable routing paths with secure blockchain-assisted operations. Significantly, the DWPO-based routing framework outperforms existing models, providing an energy-efficient, reliable, and secure solution for VANET routing. The integration of blockchain with DWPO enhances trustworthiness and overall network performance, making it suitable for real-time vehicular communication applications.

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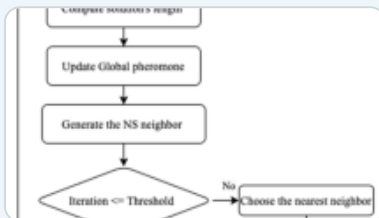
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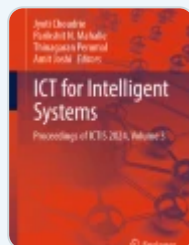
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## Data Availability

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Not applicable.

## Code Availability

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Not applicable.

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## **Ethics declarations**

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The authors declare no competing interests.

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